To retrieve events by tag, whether current or not, you must pass the tag and offset. Events have a `sequence_number` when stored in the journal; the offset is the corresponding number that marks the latest `sequence_number` read. To read from the beginning, you can pass a special object, the `Offset.noOffset`.

To read from the journal, you must first select a plugin that implements the Akka Persistence Query API. Akka provides and maintains three plugins: Cassandra, Couchbase, and JDBC. (You can find others at http://mng.bz/5wN8.)

In chapter 9, we used a database configuration for the connection to write the journal, but this was not explicitly mentioned in the chapter. Listing 10.3 shows that configuration, which can also be used to read the journal. It uses the JDBC plugin to store snapshots and journal events. The underlying storage is PostgreSQL.

NOTE The source code for this example is available at www.manning.com/books/akka-in-action-second-edition or https://github.com/franciscolopez sancho/akka-topics/tree/main/chapter10b. You can find the contents of any snippet or listing in the .scala file with the same name as the class, object, or trait.

Listing 10.3 JDBC persistence configuration

```
akka.persistence {
    journal.plugin = "jdbc-journal"                          ❶
    auto-start-journals = ["jdbc-journal"]                   ❷

    snapshot-store.plugin = "jdbc-snapshot-store"            ❸
    auto-start-snapshot-stores = ["jdbc-snapshot-store"]     ❹
}

jdbc-journal {
  use-shared-db = "default"                                  ❺
}

jdbc-snapshot-store {
  use-shared-db = "default"                                  ❻
}

akka-persistence-jdbc {
  shared-databases {
    default {                                                ❼
      profile = "slick.jdbc.PostgresProfile$"                ❽
      db {
        host = "localhost"
        url = "jdbc:postgresql://127.0.0.1:5432/containers"
        user = containers
        password = containers
        driver = "org.postgresql.Driver"
        numThreads = 5
        maxConnections = 5
        minConnections = 1
      }
    }
  }
}
```

❶ Journal plugin configuration entry used by a persistent actor by default

❷ List of journal plugins to start automatically

❸ Snapshot plugin configuration entry used by a persistent actor by default

❹ List of snapshot stores to start automatically